BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 124 · Lesson 124 · Procedural Programming: Modules and State

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Which paradigm describes a desired result, rather than detailed step-by-step changes?

Q2. A full adder receives A=1, B=1 and carry-in=0. What are sum and carry-out?

Q3. For the binary value 1010, what is its decimal value?

Q4. A linked-list node has data and what other essential field?

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I Do · Read new knowledge

Key vocabulary

procedure — Named subroutine mainly performing an action.

function — Subroutine that returns a value.

parameter — Named input used by a routine.

scope — Region where an identifier may be accessed.

New learning

Sequence and control — Statements execute in an order; selection and loops direct flow.

Procedure/function — Reusable named routines accept parameters; functions return values.

Scope and state — Local variables limit unwanted side effects; global state needs care.

Worked example

Trace FUNCTION doublePlusOne(x): result=x*2; RETURN result+1; print(doublePlusOne(4)).

1. Model answer — Parameter x=4, local result=8, return value 9, printed output 9. The local result does not become a global variable merely because the function runs. — Check: link each claim to the scenario and use precise OCR terminology.

We Do · Practise with guidance

Trace FUNCTION area(w,h): RETURN w*h; print(area(3,5)). How would a procedure differ?

What must we decide?

Identify the inputs, relevant facts and exact output or judgement required.

Work through the method

Use the middle stage of the visual to follow the process and state each intermediate result.

Shared worked answer

Arguments bind w=3 and h=5; return 15; print shows 15. A procedure is intended primarily to perform an action rather than return a result (conventions vary by language).

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Explain the benefits of using functions/procedures rather than one long main routine. [6]

Question 2

Trace FUNCTION f(n): a=n+2; RETURN a*3; OUTPUT f(2). [4]

Question 3

Explain local/global scope and give a bug caused by modifying shared global state. [8]

Exit Ticket · Answer before checking

A local variable declared inside a function is automatically accessible everywhere in the program.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Declarative programming.

Q2. Sum 0 and carry-out 1.

Q3. 10.

Q4. A link/reference to the next node (for a singly linked list).

You Do pages · Independent questions

Q1. Reuse [2], modular testing [2], readability/maintainability [2].

Q2. n=2 [1], a=4 [1], return 12 [1], output 12 [1].

Q3. Local restricted scope [2], global broader visibility [2], bug example [2], mitigation with parameters/return values [2].

Exit Ticket · Model answer

False. It is restricted by scope.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: